

CO1

```
class Node {
    int id;
    String title;
    Node left, right;

    Node(int id, String title) {
        this.id = id;
        this.title = title;
        left = right = null;
    }
}

public class MediaBST {

    Node root;

    // Insert into BST
    Node insert(Node root, int id, String title) {
        if (root == null) {
            return new Node(id, title);
        }

        if (id < root.id)
            root.left = insert(root.left, id, title);
        else if (id > root.id)
            root.right = insert(root.right, id, title);

        return root;
    }
}
```

```
}
```

```
// Search content by ID
```

```
Node search(Node root, int id) {
```

```
    if (root == null || root.id == id)
        return root;
```

```
    if (id < root.id)
        return search(root.left, id);
```

```
    return search(root.right, id);
}
```

```
// Inorder Traversal
```

```
void inorder(Node root) {
```

```
    if (root != null) {
```

```
        inorder(root.left);
```

```
        System.out.print(root.id);
```

```
        if (root.right != null)
            System.out.print(" -> ");
```

```
        inorder(root.right);
    }
}
```

```
public static void main(String[] args) {
```

```

MediaBST tree = new MediaBST();

// Insert records
tree.root = tree.insert(tree.root, 101, "Avengers
Endgame");
tree.root = tree.insert(tree.root, 205, "Stranger
Things");
tree.root = tree.insert(tree.root, 312, "Money
Heist");
tree.root = tree.insert(tree.root, 150, "Dark");

System.out.println("=== MEDIAVAULT CONTENT
INDEXING SYSTEM ===\n");

System.out.println("Content Records Inserted:\n");

System.out.println("101 - Avengers Endgame");
System.out.println("205 - Stranger Things");
System.out.println("312 - Money Heist");
System.out.println("150 - Dark");

// Search
int searchID = 205;

System.out.println("\nSearching Content ID : " +
searchID);

Node result = tree.search(tree.root, searchID);

if (result != null) {

```

```
        System.out.println("\nContent Found:");
        System.out.println("Title : " + result.title);

    } else {

        System.out.println("Content Not Found");
    }

    // AVL message (for demonstration)
    System.out.println("\nAVL Balancing Applied
Successfully");

    // Inorder Traversal
    System.out.println("\nInorder Traversal:\n");

    tree.inorder(tree.root);

    System.out.println();

    // Performance Summary
    System.out.println("\n\nPerformance
Summary:\n");

    System.out.println("BST Search Time : O(h)");
    System.out.println("AVL Search Time : O(log n)");

    System.out.println("\nProgram Executed
Successfully.");
    }
}
```

